

Discussion: “Banks & State-Dependent MonPol” by Eichenbaum, Puglisi, Rebelo & Trabandt

Benedikt Kolb (Bundesbank)¹

6th Biennial Conference NBP/LB/CEBRA, Warsaw, 25 Sep. 2026

¹benedikt.kolb@bundesbank.de. These are not necessarily Bundesbank views.

Truly interesting mechanism, very timely topic

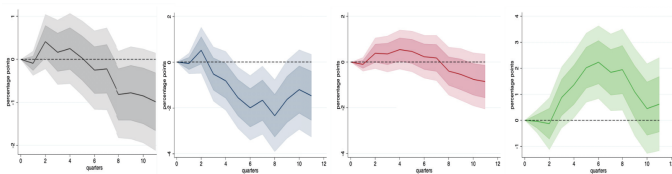
- Paper combines two important issues:
 - Effectiveness of monetary policy when rates are already high
 - (Depositor) attention in macro
- Elegant and novel model
- Clear structure, model setup is well explained
- Many details and robustness checks

Part I: Recap

Core empirical findings

- Contractionary US MonPol less powerful when rates are high.

Panel B: Log Real Per Capita GDP



(e) Baseline

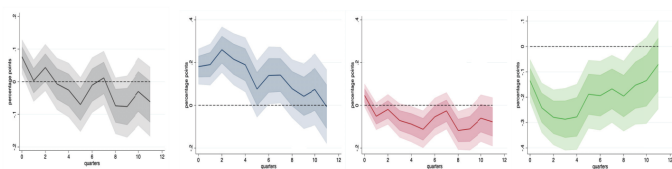
(f) Low Rate State

(g) High Rate State

(h) High vs Low

- Bank's net interest margin (NIM) *increases* when rates are low, but *decreases* when they are high.

Panel B: Core NIM



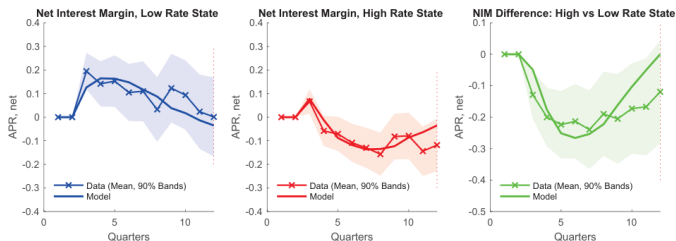
(e) Baseline

(h) High vs Low

Model to replicate NIM dynamic (partial equilibrium)

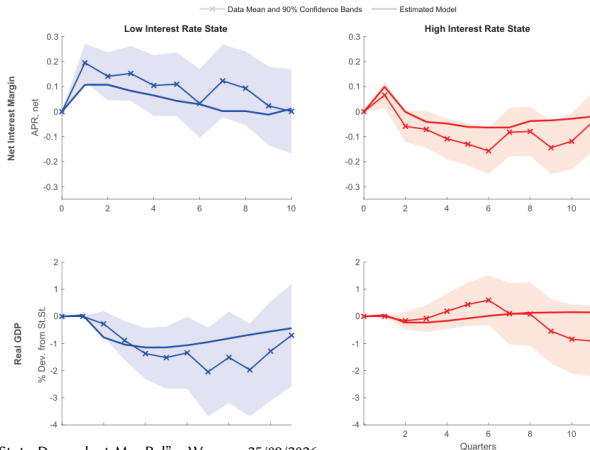
- Depositor heterogeneity: attentive and inattentive type
- Attentive types are more costly for banks (lower spread earned on them).
- “Social dynamics”: Share of attentive depositors increases when rates are high.
- This reduces the NIM of banks starting from higher rates.

Figure 6: Dynamic response to monetary policy shock: NIM



Model to replicate GDP dynamic (general equilibrium)

- Full-blown NK model with many bells and whistles
 - Tightening starting from high rates: NIM effects from PE model translate to higher deposit income of high-MPC HtM consumers
- Less effective MP tightening!



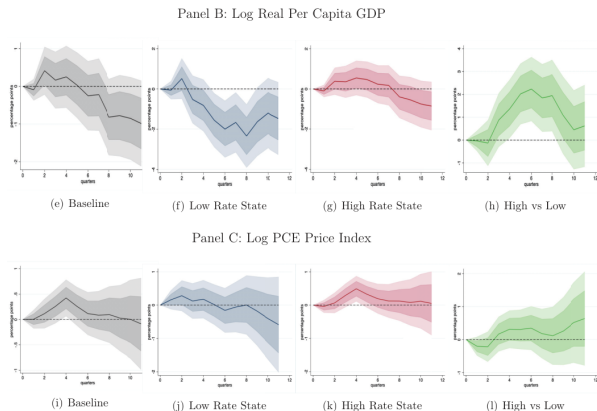
Part II: Comments

1. Tie paper to booming lit. on “attention” in macro

- (cited) Pfäuti (forthcoming AER): Public attention to inflation doubles when inflation passes an inflation rate of 4 percent.
 - Link et al. (forthcoming QJE) and Weber et al. (2025, ECTA): Attention to inflation increases with inflation level and leads to more information-gathering.
 - Related literature on memory and attention buildup: Bordalo et al. (2020), Graeber et al. (2024)
- This paper: structural model with endogenous attention, which then leads to state-dependence in monetary policy

2. Price puzzle in empirical baseline

Recursive MP shock identification often implies price puzzles w/ post-2000 data ([Barakchian and Crowe, 2013](#)). So does EPRT (Fig. 3):



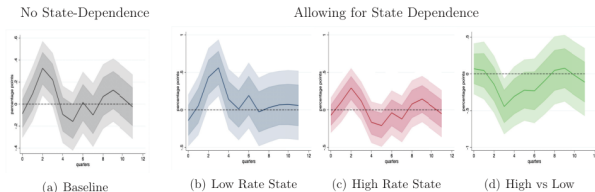
Even worse for [1984q1-2008q4 sample](#). (No puzzle in [CEE99](#) & [CEE05](#).)
→ [Bauer and Swanson \(2023\)](#) as baseline? Forward-looking controls?

3. Time and savings deposits against your story?

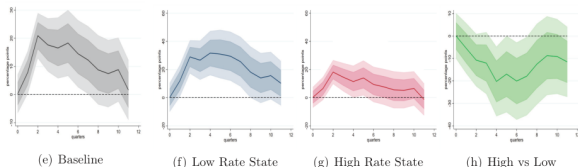
Time deposits carry higher interest than savings deposits.

→ Higher rates and volumes of time relative to savings deposits would tend to *decrease* the NIM when rates are low?

Panel A: Time Deposit Rate minus Saving Deposit Rate



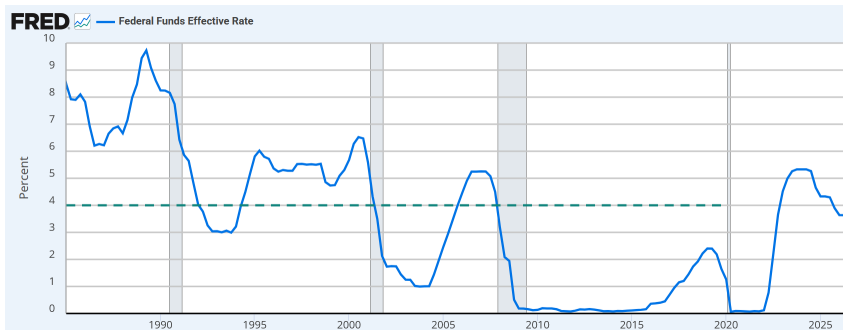
Panel B: Time Deposit Volume over Saving Deposit Volume



4. Hard 4% threshold

4% well-motivated by Pfäuti (forthcoming AER), but:

- a) High-rate episodes clearly clustered in 1980s and 90s
- b) Hard cutoff, while model suggests continuous state-dependence
→ Why not use a level interaction (or smooth transition)?



5. Headline GE numbers appear stretched

Δ cumulative GDP response: \$130bn

Paper roughly matches this by:

1. \$48 bn by stock market decline

- Δ S&P 500 response from empirical model
- BUT: S&P 500 IRF not fully significant at that horizon

2. \$60bn from NIM difference

- $\sum_1^{12} \Delta$ (NIM) response from empirical model ($\sim 150\text{bn}\$$) $\times$ “MPC of liquid wealth” estimate of 0.4 (40%)
- BUT: large end of empirical MPC range; maximum in 0.2-0.4 range in [Carroll et al. \(2017\)](#);
[Kaplan et al. \(2014\)](#): rather 0.1-0.3

KVW HtM US

6. GE model results depend on high MPC traction

- Paper: For NIM state-dependence to move demand, high-MPC HH must receive the deposit interest. The model has
 - 50% HtM HHs (posterior from IRF matching)
 - who deposit wages *within the period* and earn *intra-period* interest, while making “no intertemporal consumption choices”.
- [Kaplan et al. \(2014\)](#): ~ a third of US households are HtM. Of these households, *roughly two-thirds are wealthy*—owning an illiquid asset.

7. Allow competing explanations for state dependence?

Explanations for state-dependence in lending rates:

- [Berger et al. \(2021, AER\)](#) and [Eichenbaum et al. \(2022, AER\)](#) :
When rates are high, the share of mortgagors who choose not to refinance increases, reducing MP bite.
- [Drechsler et al. \(2017, QJE\)](#) “deposit channel of MP”: Banks widen NIM after MP tightening → deposit outflow → reduced lending
→ What if deposit-channel varies over interest rates?
- [Greenwald et al. \(2024\)](#): MP transmission “stronger when more banks are required to adjust their regulatory capital for unrealized value changes of securities.”
→ More likely after long period of low rates!

These can help to explain state-dependent GDP reaction in total.

→ Empirical part already finds an increase in loan interest income!

8. Opens avenues for further research

1. Government spending multiplier:

- Ramey and Zubairy (2018, JPE): GSM larger when interest rates are at the ZLB
 - Their shocks often have disinflationary effects (Jørgensen and Ravn, 2022).
- Coming from a low-rate environment, MP easing after the fiscal shock might be more effective

2. Financial stability:

- MP hike costly for banks above deposit costs:
 - bond portfolios losses
 - credit default risk up
 - reduced loan demand
 - On average, NIM increases might counteract this.
 - This paper: under high rates, MP tightening *decreases* NIM.
- Fin.-stability risks from MP hikes in high-rate environments?

9. Minor points

- Estimated deposit stickiness $\delta \approx 1.3\%$ per quarter $\rightarrow$ implies rather extreme deposit duration of about 19 yrs. Present-value effect scales with δ . Are results sensitive to that value?
- Comparatively high MPC measure of 0.6 from [Ganong et al. \(2024\)](#) derived from sudden high-liquidity states (unemployment benefit windfalls) and questionable for calibrating average MPCs.
- Bands assume zero correlation between β_0 and β_1 (fn. 7); but paper shows negative correlation in the data—this would imply wider bands.

Conclusion

Findings:

- Empirical: In high-rate environments, monetary policy is less effective and banks' net interest margin is state-dependent.
- Model: Depositors become more attentive under high rates. Higher deposit-rate income of high-MPC impatient HHs counteracts monetary tightening.

What I would address:

- price puzzle in empirical part
- state-dependent lending rate in model

Truly novel mechanism + timely question = high publication potential!

Appendix

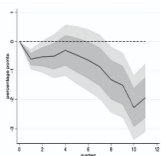
Very minor points

- Fig 1 would benefit from having the same y-axis, at least for the first three plots per line
- Fig 1: LPs find a stronger and more persistent increase in FFR for the low-rate state. That works in same direction as the NIM channel (FFR hikes will be less effective in a high-rate environment). How much of the NIM effect remains when controlling for this (e.g. rescale s.t. maximum instead of impact FFR effect is same)?
- “First, if we do not allow for state dependence, core NIM falls by a modest amount after a contractionary monetary policy shock.” - if anything, there is a significant increase on impact
- footnote 8 is confusing - is the intensive margin not the spread between time and savings deposit rates?
- IRF-matching posteriors lie mostly outside the prior distributions (good!). But: rates at which attentive and inattentive depositors switch camps are really low. Is this in line with empirics?

Price puzzle in sample 1984Q1-2008Q4

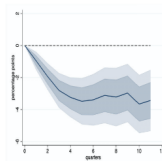
Panel A: Logarithm of real per capita GDP

No State-Dependence

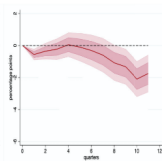


(a) Baseline

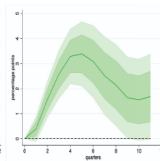
Allowing for State Dependence



(b) Low Rate State

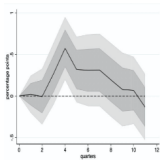


(c) High Rate State

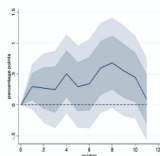


(d) High vs Low

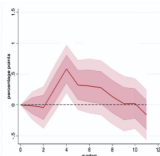
Panel B: Logarithm of PCE deflator index



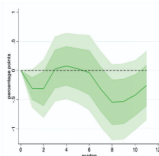
(e) Baseline



(f) Low Rate State



(g) High Rate State



(h) High vs Low

[back](#)

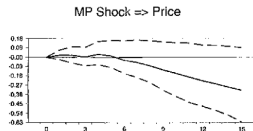
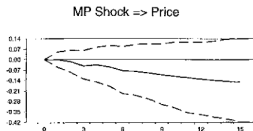
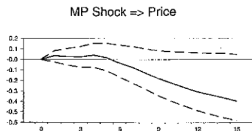
Benedikt Kolb

Disc.: "Banks & State-Dependent MonPol" – Warsaw, 25/09/2026

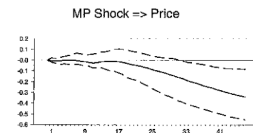
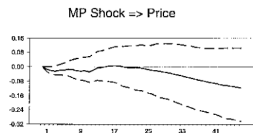
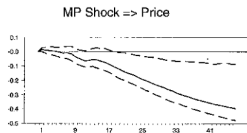
Page 2 / 10

I No price puzzle in CEE (1999) [back](#)

The VAR in [Christiano et al. \(1999\)](#) has **no** price puzzle:
CEE (1999) quarterly VAR:



CEE (1999) monthly VAR:



No price puzzle in CEE (2005) [back](#)

The VAR in [Christiano et al. \(2005\)](#) has **no** price puzzle:

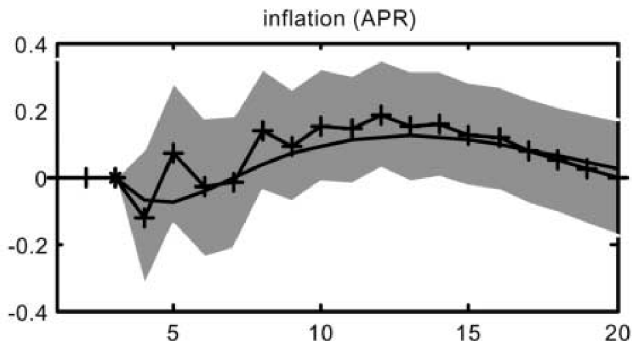


FIG. 1.—Model- and VAR-based impulse responses. Solid lines are benchmark model impulse responses; solid lines with plus signs are VAR-based impulse responses. Grey areas are 95 percent confidence intervals about VAR-based estimates. Units on the horizontal

Table 6. Marginal Propensity to Consume out of Transitory Income Shocks for Different Types of HtM Households, United States^a

	<i>P-HtM</i>	<i>W-HtM</i>	<i>N-HtM</i>	<i>HtM-NW</i>	<i>N-HtM-NW</i>
Baseline	0.243*** (0.065)	0.301*** (0.048)	0.127*** (0.036)	0.229*** (0.054)	0.201*** (0.030)
Pre-tax earnings ^b	0.131*** (0.043)	0.223*** (0.035)	0.122*** (0.027)	0.143*** (0.036)	0.164*** (0.023)
Include food stamps ^c	0.217*** (0.059)	0.264*** (0.045)	0.105*** (0.035)	0.203*** (0.050)	0.171*** (0.029)
Continuously married households ^d	0.095 (0.194)	0.193** (0.079)	0.079* (0.043)	-0.048 (0.129)	0.157*** (0.042)
Stable marital status ^e	0.239*** (0.085)	0.282*** (0.054)	0.110*** (0.038)	0.190*** (0.070)	0.195*** (0.033)
Households with male heads ^f	0.186** (0.080)	0.193*** (0.058)	0.073* (0.040)	0.150** (0.064)	0.129*** (0.035)
Monthly income ^g	0.229*** (0.068)	0.288*** (0.053)	0.159*** (0.034)	0.236*** (0.057)	0.199*** (0.030)

Source: Authors' calculations, based on United States PSID. See text for full description.

a. Boot-strapped standard errors based on 250 replications in parentheses. Statistical significance indicated at the ***1 percent; **5 percent; and *10 percent levels.

b. Transfers are excluded.

c. Food stamps are included among transfers.

d. Restricted to continuously married households.

e. Restricted to households with no change in marital status.

f. Households with female heads (mostly single) are excluded.

g. Pay period is set to one month instead of two weeks.

References I

- Barakchian, S. M. and C. Crowe (2013, None). Monetary policy matters: Evidence from new shocks data. *Journal of Monetary Economics* 60(8), 950–966.
- Bauer, M. D. and E. T. Swanson (2023, March). An alternative explanation for the “fed information effect”;. *American Economic Review* 113(3), 664–700.
- Berger, D., K. Milbradt, F. Tourre, and J. Vavra (2021, September). Mortgage prepayment and path-dependent effects of monetary policy. *American Economic Review* 111(9), 2829–2878.
- Bordalo, P., N. Gennaioli, and A. Shleifer (2020, None). Memory, attention, and choice. *The Quarterly Journal of Economics* 135(3), 1399–1442.

References II

- Carroll, C., J. Slacalek, K. Tokuoka, and M. N. White (2017, November). The distribution of wealth and the marginal propensity to consume. *Quantitative Economics* 8(3), 977–1020.
- Christiano, L. J., M. Eichenbaum, and C. L. Evans (1999, None). Monetary policy shocks: What have we learned and to what end?
- Christiano, L. J., M. Eichenbaum, and C. L. Evans (2005, February). Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy* 113(1), 1–45.
- Drechsler, I., A. Savov, and P. Schnabl (2017, None). The deposits channel of monetary policy. *The Quarterly Journal of Economics* 132(4), 1819–1876.

References III

- Eichenbaum, M., S. Rebelo, and A. Wong (2022, March). State-dependent effects of monetary policy: The refinancing channel. *American Economic Review* 112(3), 721–761.
- Ganong, P., F. Greig, P. Noel, D. M. Sullivan, and J. Vavra (2024, September). Spending and job-finding impacts of expanded unemployment benefits: Evidence from administrative micro data. *American Economic Review* 114(9), 2898–2939.
- Graeber, T., C. Roth, and F. Zimmermann (2024, None). Stories, statistics, and memory. *The Quarterly Journal of Economics* 139(4), 2181–2225.
- Greenwald, D. L., J. Krainer, and P. Paul (2024, May). Monetary transmission through bank securities portfolios. Working Paper Series 2023-18, Federal Reserve Bank of San Francisco.

References IV

- Jørgensen, P. L. and S. H. Ravn (2022, None). The inflation response to government spending shocks: A fiscal price puzzle? *European Economic Review* 141(C), None.
- Kaplan, G., G. L. Violante, and J. Weidner (2014, None). The wealthy hand-to-mouth. *Brookings Papers on Economic Activity* 45(1 (Spring)), 77–153.
- Link, S., A. Peichl, O. Pfäuti, C. Roth, and J. Wohlfart (forthcoming). Attention to the macroeconomy. *Quarterly Journal of Economics*.
- Pfäuti, O. (forthcoming). The inflation attention threshold and inflation surges. *American Economic Review*.
- Ramey, V. A. and S. Zubairy (2018, None). Government spending multipliers in good times and in bad: Evidence from us historical data. *Journal of Political Economy* 126(2), 850–901.

References V

Weber, M., B. Candia, H. Afrouzi, T. Ropele, R. Lluberas, S. Frache, B. Meyer, S. Kumar, Y. Gorodnichenko, D. Georgarakos, O. Coibion, and Geoff (2025, January). Tell me something i don't already know: Learning in low and high inflation settings. *Econometrica* 93(1), 229–264.